This month we highlight some aspiring tech writers and their spirited attempt to get published in the magazine. Let's have more!

Readers' write-ups

Tarun Dham

Technology In Sports

Technology is an integral part of our lives. Most of the things we use currently wouldn't have existed without technology. Even sports isn't unaffected. Let's take a look at some technology specifically for sports:

CRICKET

People are crazy about cricket in India, but they hardly have a clue about the technology used in the sport. Obviously, the looking-glass that is your TV is also an important technological creation.

Snickometer

Created by Allan Plaskett, this technique helps umpires decide whether a batsman is out or not out. A sensitive microphone is kept near the stumps and it's connected to an oscilloscope that measures sound waves made by the ball touching the bat. On contact, the oscilloscope responds with a big wave (of sound). The decision is made when the wave is synchronised and played with the slow motion shot.

Hawkeye

This brilliant piece of technology was created by Dr. Paul Hawkins. It uses sensitive image processing techniques using cameras that are placed all over the stadium which track the trajectory of the ball thrown by the bowler. This helps accurately track the ball's speed and length, providing the decision that is used in LBW. The Hawkeye is not only used in cricket but also in tennis as well, to help resolve disagreements.

Dartfish

Dartfish is used by team coaches to analyse a player's performance, statistics etc. It not only shows the player's live footage, but also compares the footage to the last-played match. This sophisticated system breaks the recorded footage into frame-by-frame sequences for the actions of each player, helping performance analyses.

FORMULA 1

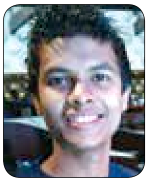
Everyone is familiar with this sport which pits professional drivers against each other for a race using super fast F1 cars at break-neck speeds. Since the pace of this sport is extremely quick, it is continuously monitored by the race teams, and the racer is given second-by-second reports. There is a lot of impressive technology used :

Car Design

A regular F1 car is capable of going from 0 to 160km/h in less than five seconds and the design of the car matters a lot. To handle this, each and every F1 car is meticulously designed. The chassis of a typical F1 car is made of composite carbon fibre and lightweight materials and thus the weight of the car is very less but ballast is placed inside the car to improve weight distribution. This helps lower the car's centre of gravity, thus increasing its stability. The car is designed keeping aerodynamics in mind. Good aerodynamics minimise drag, lowering turbulence. The wings of an F1 car are very important too as they help in providing maximum downforce.

Steering wheel

Using the steering wheel of the F1 car, a driver has the ability to control many elements of the car. The buttons and knobs on the steering wheel can be used to change gears, adjust fuel/air mix, apply rev. limiter, change brake pressure, use the radio. A small LCD screen on the steering wheel displays data such as speed, engine RPM, lap times, and the gear. The steering wheel has gear shift paddles as well.



Tanmay Patange

My visit to the YouTube API Developers Conclave

A big high five to Digit Squad members and readers alike who share the knowlegde they obtain from this magazine. I would like to share one of my experiences with all you readers. Earlier this month, I attended the YouTube API Developers Conclave in Mumbai. The turnout was impressive and the attendees' response was great All of them looked very passionate about YouTube APIs, especially for application development.

At the venue, I met Jonathan Levine, Associate Product Manager at Google. Here are some of the key points I discussed:

Me: I recently saw a movie called "The Internship", which was really fascinating. It shows the interns going through some tough challenges which they had never imagined. Is the process actually that difficult or is that just the Movie?

Jonathan: (Laughs) Not really! I agree that the process is not so easy to complete. However, whatever is shown in the movie is not 100% real, after all it's just a film right?

Me: Yes, exactly. Moreover, is it really the Google HQ shown in the movie?

Jonathan: Yes, at some parts. Some of the movie scenes were filmed inside Google headquarters, and I remember it happening just outside my cabin. It was really a great experience for all of us at Google.

Me: Google introduced the Chromebook Pixel and the Chrome OS. Is there any reason why Google employees are usually seen working on a MacBook Pro?

Jonathan: (Laughs again) Maybe yes! I have Chromebook Pixel, and in terms of hardware specifications, it simply stands